ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	The properties of the convex mirror explain its use.
2	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R does not explain why the image is inverted.
3	**(a)** Both A and R are true and R is the correct explanation of A.	High resistance draws almost no current.
4	**(a)** Both A and R are true and R is the correct explanation of A.	At a crossing there would be two directions of the field.
5	**(a)** 11.25 cm	1/f = -1/45 - 1/15 = -4/45, so f = -11.25 cm.
6	**(b)** -25 cm	f = -(far point) for objects at infinity.
7	**(d)** 6 ohm	R2 = rho 2L / 4A = R / 2.
8	**(a)** 90 degrees	Perpendicular is the maximum.
9	**(a)** convex, f = 12 cm	Object at 2F.
10	**(b)** smaller wavelengths are scattered more by small particles	Rayleigh type scattering.
11	**(a)** V against I	Ohm's law.
12	**(d)** 100 Hz	Proportional to speed.
13	**(a)** Both A and R are true and R is the correct explanation of A.	The wide field of view of the diminished, erect image lets the driver see more traffic behind.
14	**(a)** Both A and R are true and R is the correct explanation of A.	In myopia the image forms in front of the retina; the diverging lens shifts it back onto the retina.
15	**(a)** Both A and R are true and R is the correct explanation of A.	A low-resistance meter in series does not change the current it is measuring.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	m = -1/3 = -v/u, v = -15, so u = -45 cm (1); 1/f = -1/15 - 1/45 = -4/45, f = -11.25 cm (1).	1+1
17	Red is scattered the least (1); The scattered light is dominated by shorter wavelengths (1).	1+1
18	I = 4.8×10^{-3} A (1); n = 4.8×10^{-3} / 1.6×10^{-19} = 3×10^{16} (1).	1+1
19	Force is perpendicular to the velocity (1); No work is done, so the speed is constant (1).	1+1
20	Image of the Sun is at the focus, f = 20 cm (1); R = 2f = 40 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	First case: u = -x, v = -3x, so 1/f = -1/(3x) - 1/x = -4/(3x), |f| = 3x/4 (1); Second case: same size means the object is at C, so x + 10 = 2|f| = 3x/2 (1); x = 20 cm, so f = -15 cm (1).	1+1+1
22	1/f = -1/30 + 1/25 (1); 1/f = 1/150, f = 150 cm (1); P = +0.67 D (1).	1+1+1
23	Atoms vibrate more, increasing collisions with electrons (1); Resistance increases (1); Current decreases (I = V / R) (1).	1+1+1
24	Force arises from the component of the field perpendicular to the wire (1); It is maximum when the wire is perpendicular (1); It is zero when the wire is parallel (1).	1+1+1
	OR: m = -5, v = -5u (1); 1/12 = 1/(5|u|) - ... with u negative: 1/12 = 1/(5|u|) + 1/|u| = 6/(5|u|) (1); |u| = 14.4 cm, v = 72 cm (1).	1+1+1
25	Light reaches us after passing through a thick layer of air (1); Light is scattered many times by particles of all sizes (1); Colours mix and appear whitish (1).	1+1+1

	OR: P = 300 W (1); I = 300 / 220 = 1.36 A (1); R = 220 / 1.36 = 161.3 ohm (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $f = R/2 = -20$ cm (1). (ii) $u = -30$, $f = -20$: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = -\frac{1}{20} + \frac{1}{30} = -\frac{1}{60}$, so $v = -60$ cm, i.e. 60 cm in front of the mirror (1). (iii) $m = -\frac{v}{u} = -\frac{-60}{-30} = -2$ (1). (iv) Image height $= 2 \times 3 = 6$ cm (1); real, inverted and enlarged (1).	1+1+1+2
27	(a) Parallel part = 48 / 16 = 3 ohm; total = 9 ohm (1); (b) I = 12 / 9 = 1.33 A (1); (c) Voltage across parallel = 1.33 x 3 = 4 V (1); Currents: 4/12 = 0.33 A and 4/4 = 1 A (1); (d) P = 12 x 1.33 = 16 W (1).	1+1+1+1+1
	OR: Fleming's left-hand rule (1); Forefinger north, middle finger east (1); Thumb points upward (1); (a) Force becomes downward (1); (b) Both reversed: force upward as before (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the deflection is evidence of the field, not its explanation.
2	**(b)** -40 cm	m = -v/u = 2 gives v = +40; 1/f = 1/40 - 1/20 = -1/40.
3	**(b)** -1 D and +1.5 D	Distance: f = -100 cm. Reading: 1/f = -1/40 + 1/25 = 3/200, P = +1.5 D.
4	**(b)** 9 R	L is tripled, A is one third.
5	**(b)** zero	No change of linked field.
6	**(b)** 1.5 f	1/v = 1/f - 1/3f = 2/3f.
7	**(c)** blue light more strongly than red light	Shorter wavelengths are scattered more.
8	**(a)** 5 ohm	R = V / I.
9	**(b)** 3 E	Proportional to rate of change.
10	**(d)** A is false but R is true.	A is false (image is virtual); R is true.
11	**(a)** Both A and R are true and R is the correct explanation of A.	Higher n means more bending.
12	**(a)** Both A and R are true and R is the correct explanation of A.	Greater collisions give more resistance.
13	**(a)** Both A and R are true and R is the correct explanation of A.	If two lines crossed, there would be two directions of the field at the point of intersection.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the ray falls normally on the mirror and retraces its path, which is not explained by $R = 2f$.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R describes how the rainbow forms but does not explain the required geometry of Sun, observer and rain.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Colours always separate (1); Because n differs for each colour (1).	1+1
17	R = 220^2 / 800 = 60.5 ohm (1); H = P t = 800 x 300 = 2.4×10^5 J (1).	1+1
18	I = 2200 / 220 = 10 A (1); The 15 A circuit is suitable (1).	1+1
19	Ray hits perpendicular to the surface (1); No bending and no sideways displacement (1).	1+1
20	Iris limits the entry of light (1); Lens changes its curvature to focus (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Metal case can become live due to a fault (1); The earth wire is a low-resistance path (1); The fuse blows and the user is safe (1).	1+1+1
22	Water is rarer than glass, so the ray bends away from the normal (1); n(w,g) = 1.33/1.5 (1); = 0.887 (1).	1+1+1
23	At 10 cm (1); The lens must be thicker and have a shorter focal length (1); For 100 cm the lens is thinner (1).	1+1+1
24	Ammeter in series should not add to the resistance (1); Voltmeter in parallel should not draw current (1); Else the readings are altered (1).	1+1+1
	OR: Current produces a magnetic field (Oersted) (1); Changing magnetic field produces current (Faraday) (1); Force on a conductor in a field links the two (1).	1+1+1
25	m = v/u = -3, v = 80, u = -26.7 cm (1); 1/f = 1/80 + 1/26.7 (1); f = 20 cm (1).	1+1+1
	OR: Hypermetropia and myopia (1); Reading: 1/f = -1/100 + 1/25, P = +3 D (1); Distance: f = -2 m, P = -0.5 D (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Myopia (short-sightedness) (1). (ii) Concave (diverging) lens (1). (iii) Excessive curvature of the eye lens, or elongation of the eyeball (1). (iv) The lens must form the image of a distant object at the far point: $f = -80$ cm $= -0.8$ m (1); $P = \frac{1}{f} = \frac{1}{-0.8} = -1.25$ D (1).	1+1+1+2
27	20 ohm: P = 14400 / 20 = 720 W (1); 30 ohm: P = 480 W (1); Series 50 ohm: 288 W (1); Parallel 12 ohm: 1200 W (1); Parallel gives the highest power (1).	1+1+1+1+1
	OR: Both have similar field patterns (1); Lines go from N to S outside (1); The solenoid's field can be switched on or off (1); A soft iron core makes it a strong electromagnet (1); Scrap is lifted while the current flows and dropped when switched off (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** R / n^2	Each part R/n; parallel gives R/n^2.
2	**(b)** develops an induced current while the magnet moves	A changing field links the coil only during the motion.
3	**(a)** +10 cm	Convex magnifier: m = 4, v = -30, u = -7.5; 1/f = -1/30 + 1/7.5 = 1/10.
4	**(b)** larger than the wavelength of light	All colours scatter.
5	**(b)** 1.5 A	R = 2 + 2 = 4 ohm.
6	**(c)** zero	Positive and negative halves cancel.
7	**(a)** Both A and R are true and R is the correct explanation of A.	A thin parallel-sided region gives no net deviation.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Shorter wavelength scatters more.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Parallel arrangement features.
10	**(d)** A is false but R is true.	A is false: the force is zero when the conductor is parallel to the field; R is true.
11	**(d)** 2	n = 3/1.5.
12	**(a)** white light	The second prism reverses the dispersion.
13	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true ($R = \rho l/A$ and ρ changes with temperature), but R does not explain A.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the field pattern outside is like that of a bar magnet, which is not explained by the uniform field inside.
15	**(c)** A is true but R is false.	Glass is optically denser: light slows down and bends towards the normal; R is false.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	R = V^2 / P (1); Ratio R100 : R25 = 1 : 4 (1).	1+1
17	It disconnects the appliance from the high potential (1); The appliance stays safe when switched off (1).	1+1
18	Lens: 1/v = 1/20 - 1/40, v = +40 cm (1); Mirror: 1/v = -1/20 + 1/40, v = -40 cm (1).	1+1
19	1/f = -1/50 + 1/25 = 1/50 (1); P = +2 D (1).	1+1
20	Two halves of 2 ohm each in parallel (1); R = 1 ohm (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Area ratio = 1 : 4 (1); R1 : R2 = 4 : 1 (1); Parallel: I1 : I2 = 1 : 4 (1).	1+1+1
22	Inside: field lines are nearly parallel and in one direction (1); Outside: lines go back in the opposite direction (1); Each element of the wire adds to the field (1).	1+1+1
23	1/v = 1/2 + 1/4 = 3/4, v = +1.33 m (1); m = -v/u = 1.33/4 (1); m = 0.33 (image is one-third) (1).	1+1+1
24	Prism disperses the light (1); Edges appear coloured due to unequal deviations of colours (1); Violet and red are separated (1).	1+1+1
	OR: Voltmeter has a very high resistance (1); Total resistance becomes very high (1); Almost no current flows and the bulb will not glow (1).	1+1+1
25	Straight wire: concentric closed circles around the wire (1); Bar magnet: lines emerge from N and enter S (1); Both have closed loop fields (1).	1+1+1
	OR: Angle of incidence is the angle with the normal = 0 (1); Angle of reflection = 0 (1); The ray retraces its path (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $R = V/I = 2/0.4 = 5\ \Omega$ (1). (ii) A straight line passing through the origin (1). (iii) At constant temperature the current through a conductor is directly proportional to the potential difference across its ends (1). (iv) Each half has resistance $2.5\ \Omega$ (1); in parallel $R = 2.5/2 = 1.25\ \Omega$ (1).	1+1+1+2
27	New R = 9 x 10 = 90 ohm (1); Each third = 30 ohm (1); 1/R = 3 / 30 (1); R = 10 ohm (1); Resistance comes back to the original (1).	1+1+1+1+1
	OR: A rectangular coil is placed between the poles of a magnet (1); Current in the coil gives forces on its two arms in opposite directions (1); The coil rotates (Fleming's left-hand rule) (1); Split rings reverse the current every half rotation (1); So the coil continues to rotate in the same direction (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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